MAY UNIVERSITY IN CAIRO

Fall 2025/2026

Sheet 1

FACULTY OF ENGINEERING

Program : Computer Engineering Program Date : **Week 1**

Examiners : Dr. **Amaal Samir**

Course/Code : **Artificial Intelligence and Intelligent Agents / CSE332** Total Marks : 10 Points

Q1. Multiple Choice Questions (MCQs)

1. Intelligence can best be defined as:

   A) The ability to act automatically
   B) The ability to understand, learn, solve problems, and make decisions
   C) The ability to store large amounts of data
   D) The ability to compute very fast

   Answer: B

2. The main goal of Artificial Intelligence is:

   A) To replace humans completely
   B) To create robots only
   C) To make machines perform tasks requiring intelligence when done by humans
   D) To increase computer speed

   Answer: C

3. The Turing Test was proposed by:

   A) John McCarthy
   B) Marvin Minsky
   C) Alan Turing
   D) Claude Shannon

   Answer: C

4. In Phase 2 of the Turing Imitation Game:

   A) Two humans try to fool a computer
   B) A computer replaces the man and tries to fool the interrogator
   C) Two computers compete
   D) The interrogator is replaced by a machine

   Answer: B

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5. Claude Shannon demonstrated the need for heuristics using:

   A) Neural networks
   B) Machine translation
   C) Chess-playing machines
   D) Expert systems

   Answer: C

6. The General Problem Solver (GPS) was developed by:

   A) Turing and Shannon
   B) Newell and Simon
   C) McCarthy and Minsky
   D) Rosenblatt and Pitts

   Answer: B

7. GPS failed mainly because:

   A) It had too much domain knowledge
   B) It required too little memory
   C) It generated too many possible operators and required excessive time and memory
   D) It was too specialized

   Answer: C

8. Early AI methods that used general search with weak domain knowledge are called:

   A) Strong methods
   B) Weak methods
   C) Probabilistic methods
   D) Neural methods

   Answer: B

Sec 1	CLO1	CLO2	CLO3	CLO4	CLO5	CLO6	CLO7	CLO8	CLO9	CLO10	CLO11	CLO12
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9. Fuzzy logic was introduced by:

A) John von Neumann
B) Alan Turing
C) Lotfi Zadeh
D) Frank Rosenblatt

Answer: C

10. A linguistic variable in fuzzy logic:

A) Has only numerical values
B) Has only binary values
C) Has values expressed as words
D) Is used only in neural networks

Answer: C

11. The Dartmouth workshop in 1956:

A) Ended AI research
B) Introduced fuzzy logic
C) Marked the birth of AI as a science
D) Introduced machine translation

Answer: C

12. The perceptron convergence theorem was proved by:

A) McCulloch
B) Rosenblatt
C) Shannon
D) Simon

Answer: B

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13. Machine translation projects in the 1960s failed mainly because:

A) Computers were too small
B) Translation requires general understanding of the subject
C) Dictionaries were unavailable
D) Russian language was too complex

Answer: B

14. The technology that deals with vague and imprecise knowledge is:

A) Bayesian networks
B) Neural networks
C) Fuzzy logic
D) GPS

Answer: C

15. The imitation game tests:

A) Speed
B) Memory
C) Intelligence
D) Vision

Answer: C

16. Fuzzy logic deals with:

A) Exact data
B) Binary values
C) Uncertain knowledge
D) Hardware

Answer: C

Sec 1	CLO1	CLO2	CLO3	CLO4	CLO5	CLO6	CLO7	CLO8	CLO9	CLO10	CLO11	CLO12
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17. Linguistic variables use:

A) Numbers
B) Symbols
C) Words
D) Codes

Answer: C

18. Intelligence is defined as the ability to:

A) Act automatically
B) Follow instincts
C) Learn, understand, solve problems, and make decisions
D) Perform calculations

Answer: C

19. The Turing Test evaluates intelligence through:

A) Visual perception
B) Physical movement
C) Communication via terminals
D) Mathematical ability

Answer: C

20. Who introduced the imitation game?

A) John McCarthy
B) Alan Turing
C) Marvin Minsky
D) Claude Shannon

Answer: B

Sec 1	CLO1	CLO2	CLO3	CLO4	CLO5	CLO6	CLO7	CLO8	CLO9	CLO10	CLO11	CLO12
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21. In phase two of the imitation game, who replaces the man?

A) Another woman
B) A robot
C) A computer program
D) An interrogator

Answer: C

22. Who is considered the second founding father of AI?

A) John von Neumann
B) Claude Shannon
C) Warren McCulloch
D) Marvin Minsky

Answer: C

23. The perceptron convergence theorem was proved by:

A) John McCarthy
B) Frank Rosenblatt
C) Alan Turing
D) Herbert Simon

Answer: B

24. Expert systems focus on:

A) Broad problem domains
B) Narrow areas of expertise
C) General intelligence
D) Random search

Answer: B

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25. Which factor mainly caused the abandonment of the General Problem Solver (GPS)?

A) Lack of funding
B) Poor user interface
C) Excessive computational time and memory requirements
D) Absence of human interaction

Answer: C

26. Claude Shannon demonstrated the necessity of heuristics in AI by analyzing:

A) Language translation
B) Neural networks
C) Chess-playing machines
D) Expert systems

Answer: C

27. The term weak methods in early AI refers to:

A) Probabilistic reasoning
B) Domain-specific expert knowledge
C) General-purpose search with little domain knowledge
D) Neural-based learning algorithms

Answer: C

28. Which limitation caused early machine translation projects to fail?

A) Limited memory
B) Lack of processing speed
C) Absence of general subject understanding
D) Poor dictionaries

Answer: C

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29. What is the core concept behind fuzzy logic?

A) Probability theory
B) Boolean logic
C) Linguistic variables
D) Rule elimination

Answer: C

30. Which feature distinguishes fuzzy expert systems from conventional expert systems?

A) Use of crisp boundaries
B) Use of binary logic
C) Faster processing with fewer merged rules
D) Elimination of expert rules

Answer: C

Q2. True or False Questions

1) Intelligence means doing things automatically without thinking.

False

2) The Turing Test provides an objective behavioral test of intelligence.

True

3) Early AI programs contained large amounts of domain knowledge.

False

4) The Dartmouth workshop officially marked the beginning of AI as a field.

True

5) GPS was successful in solving complex real-world problems.

False

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6) Weak methods use strong domain-specific knowledge.

False

7) Fuzzy logic allows computing with words rather than only numbers.

True

8) Fuzzy expert systems require no testing or tuning of rules.

False

9) Neural fuzzy systems can adjust rules based on data.

True

10) One reason for AI funding cuts in the 1970s was unfulfilled promises.

True

11) GPS was abandoned mainly due to insufficient expert knowledge.

False

12) Weak methods rely heavily on domain-specific knowledge.

False

13) Fuzzy logic models human reasoning using precise numerical values.

False

14) Linguistic variables use words instead of numbers.

True

15) Modern AI favors probabilistic and statistical methods.

True

16) Neural networks existed before expert systems.

True

17) Dartmouth workshop gave birth to AI as a science.

True

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18) Expert systems rely on strong domain knowledge.

True

19) Fuzzy logic uses crisp boundaries.

False

20) Expert systems focus on narrow domains.

True

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